VES Institute of Technology, Mumbai
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Cryptocurrency Blockchain Development - Tutorial (2025-26) / Sem 6

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1. A smart contract execution consumes 85,000 gas units. Gas price = 40 Gwei.
   If 1 ETH = $2500, find the transaction fee in ETH and USD.
   Hint : 1 gwei = 10^{-9} ETH

Solution: Total Gas Fee (ETH) = $85{,}000 \times 40 \times 10^{-9} = 3.4 \times 10^{-3}$ ETH
Fee in USD = $3.4 \times 10^{-3} \times 2500 = \$8.50$

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2. A blockchain network adjusts difficulty based on the time to mine 2016 blocks. Target time = 2 weeks = 14 days. If miners took 10 days to mine 2016 blocks, find the new difficulty. Given old difficulty = 1.0×10^{12}. Also, the formula to compute Difficulty is as follows:

$$\text{New Difficulty} = \text{Old Difficulty} \times \frac{\text{Actual Time}}{\text{Target Time}}$$

Solution:

$$\text{New Difficulty} = 1.0 \times 10^{12} \times \frac{10}{14} = 0.714 \times 10^{12}$$

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3. T1, T2, T3, T4 and T5 and the Transactions in the blockchain. Design the Merkle Tree. If transaction T3 is altered after the block is created, how would this affect the Merkle Root? How does this help maintain blockchain data integrity?

Solution:

Design a Merkle Tree for 5 transactions: T1, T2, T3, T4, and T5.

1. Hash each transaction: Calculate the hash of each individual transaction — H1 = hash(T1), H2 = hash(T2), H3 = hash(T3), H4 = hash(T4), and H5 = hash(T5).
2. Pair and hash leaf nodes: Since we have an odd number of transactions (5), duplicate the last hash H5 to make it even.
   Pairs to hash at first level:
   - Pair 1: H1 and H2 → H12 = hash(H1 + H2)
   - Pair 2: H3 and H4 → H34 = hash(H3 + H4)
   - Pair 3: H5 and H5 → H55 = hash(H5 + H5)
3. Next level: Now, hash pairs of these new hashes:
   - Pair A: H12 and H34 → H1234 = hash(H12 + H34)
   - Pair B: H55 duplicated again to make an even number → H5555 = hash(H55 + H55)
4. Root hash: Finally, hash H1234 and H5555 to get the Merkle Root:
   - Merkle Root = hash(H1234 + H5555)

Analysis of Change:

1. Change in T3's Hash: If transaction T3 changes, its hash $H_3 = hash(T3)$ changes to a new hash $H_3^{'}$.

2. Propagation of Changes Up the Tree: This altered hash changes the combined hash of its pair (here with T4's hash H_4) to $H'_{34} = hash(H'_3 + H_4)$.
3. Effect on Higher-Level Hashes: Since H'_{34} changed, any parent nodes hashing H_{34} will also change. For example, the combined hash at the next level involving H_{34} will become a different hash H'_{1234}.
4. Changes Reach the Merkle Root: Eventually, the Merkle Root hash changes to a new value $MerkleRoot'$.

Data Integrity

- **Tampering Detection:** Because the Merkle Root uniquely summarizes all transactions in the block, any change—even in a single transaction like T3—changes the root hash. This difference signals tampering.
- **Efficient Verification:** Verifiers don't need to check all transactions again but can use a **Merkle Proof** (a path of hashes from the changed leaf up to the root) to confirm integrity quickly.
- **Immutable Chain:** Blocks reference previous blocks' Merkle Roots in their headers, so any tampering in prior blocks breaks the chain's validity, ensuring strong data integrity across the blockchain.

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4. A wallet has 2 ETH initially.Find the final balance of the wallet after the following transactions : Sent 0.4 ETH; Received 1.2 ETH; Paid gas fee of 0.002 ETH.

Solution: Balance = 2 − 0.4 + 1.2 − 0.002 = **2.798 ETH**

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5. Given the Initial wallet = 5 ETH; Tx1: Send 1.5 ETH (gas 0.003); Tx2: Receive 0.8 ETH ; Tx3: Send 0.5 ETH (gas 0.002). Compute the Final Balance

Solution : Final balance = 5 − 1.5 − 0.003 + 0.8 − 0.5 − 0.002 = **3.795 ETH**

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6. A user with 0.1 BTC sends 0.04 BTC to A and 0.05 BTC to B. The transaction fee is 0.0005 BTC. Find how much change (if any) is returned.

Solution : Total sent = 0.04 + 0.05 + 0.0005 = 0.0905 BTC
Change = 0.1 − 0.0905 = **0.0095 BTC**

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7. Alice has 3.2 ETH. She sends 1.1 ETH to Bob and pays gas of 0.002 ETH. Later receives 0.6 ETH from Carol. Find her final balance.

Solution: Balance = 3.2 − 1.1 − 0.002 + 0.6 = **2.698 ETH**

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8. A Solidity function execution consumes gas units for Base transaction: 25,000; Data input: (400 bytes × 12 gas each); Storage update: 32,000. If the gas price = 45 Gwei, and 1 Gwei = 10^{-9} ETH, find the total gas fee in ETH . *(Assume 1 ETH = $2800.)*

Solution:

Total gas = 25,000 + (400×12) + 32,000
= 25,000 + 4,800 + 32,000
= 61,800

Convert gas price to ETH: 45 Gwei = 45×10^{-9} ETH per gas

Total gas fee in ETH = $61{,}800 \times 45 \times 10^{-9}$
= 2.781×10^{-3} ETH

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9. A simplified hash function outputs values from 0 to 9999. The mining rule: **hash < 500**. If the hash function is **hash = (txid + nonce) mod 10000**, and txid = 7654, find the smallest nonce that satisfies the condition.

Solution: Its a trial & Error Method

- For nonce = 346 :
  - hash = (7654+346) % 10000
  - hash **!<** 500 ⇒ **nonce = 346 is not a golden nonce**
- For nonce = 2346
  - hash = (7654+2346) % 10000 = 0
  - hash = 0 ⇒ **nonce = 2346 is a golden nonce**

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10. A simplified mining algorithm uses **hash=(txid+nonce) mod 10000.** A block is valid iff: **hash < 350.** If **txid = 9675**, determine
(a) **minimum nonce** satisfying the condition; (b) **resulting hash value**.

Solution:

- For nonce = 10000 - 9675 = 325
  - hash = (9675+325) % 10000 = 0
  - hash = 0 ⇒ **nonce = 325 is a golden nonce**

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11. A blockchain network has **6 nodes** validating transactions. If **4 nodes must approve** a transaction for consensus, determine whether consensus is achieved if only **3 nodes approve**.

Solution:

Given: Total validating nodes = 6; Required approvals = 4; Actual approvals = 3
Condition for consensus: Consensus achieved if approvals≥4
Consensus **is NOT achieved** because only **3 nodes are approved**, while **4 approvals are required**.

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12. A miner solves a hash puzzle requiring **2^{18} attempts on average**. If the miner can compute **500,000 hashes/sec**, estimate the expected time to solve the puzzle.

Solution:

Average attempts required: **2^{18}** = 262144
Hashing speed: 500000 hashes/sec

$$\text{Time} = \frac{\text{Total Attempts}}{\text{Hashes per second}}$$

$$\text{Time} = \frac{262144}{500000}$$

$$\text{Time} = 0.524288 \text{ seconds}$$

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13. A block contains **150 transactions**, each of size **250 bytes**. Calculate the **total block size** in KB.

Solution:

Transactions per block = 150
Transaction size = 250 bytes
Total size = 150×250 = 37500 bytes
We know, **1 KB=1024 bytes**

$$\text{Block Size} = \frac{\text{Total Size}}{1 \text{ KB}} = \frac{37500}{1024} = 36.62 \text{ KB}$$

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14. If a blockchain adds a new block every **10 minutes**, calculate the **number of blocks generated in one day**.

Solution:

1 block every **10 minutes**
Minutes per day = 24×60=1440 minutes

$$\text{Blocks per day} = \frac{\text{Minutes per day}}{10 \text{ minutes}} = \frac{1440}{10} = \textbf{144 blocks per day}$$

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15. A miner receives **6.25 BTC per block**. If the miner mines **3 blocks per day**, calculate the daily reward.

Solution:

Reward per block = 6.25 BTC
Blocks mined per day = 3
Daily reward =6.25×3 = **18.75 BTC**

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Extra Questions for practice

1. A blockchain block contains **200 transactions**, each requiring **21000 gas units**. Calculate the **total gas used**.
2. If the **gas price is 50 Gwei**, calculate the **transaction fee** in ETH for one transaction requiring **21000 gas**.
3. If a node stores **1 MB per block** and the blockchain has **800,000 blocks**, calculate total storage required.
4. A blockchain network has **10 nodes** where **7 nodes must approve a block**. What percentage consensus is required?
5. If Ethereum processes **15 transactions per second**, calculate the number of transactions processed in **1 hour**.
6. A mining pool contributes **10% of the network hash power**. If **144 blocks are mined daily**, estimate the number of blocks mined by the pool.
7. A miner earns **3 BTC per day**, and the price of BTC is **$30,000**. Calculate the daily earnings in USD.
8. A user sends **0.5 ETH** with a transaction fee of **0.002 ETH**. Calculate the total deducted from the wallet.
9. A cold wallet stores **5 BTC**. If BTC price rises from **$25,000 to $35,000**, calculate the portfolio value increase.
10. A multi-signature wallet requires **3 out of 5 signatures**. If only **2 users sign**, will the transaction execute?

Happy Learning!!!

V.e.S.
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